Hence for the critical value of $\gamma = 2$, the leading order correction will be zero, giving

$$\langle \vartheta \rangle_h^{[\![\mathrm{BOA}]\!]} = \langle \vartheta \rangle + \mathcal{O}(h^2).$$

This is exactly the behavior we see in Fig. 7.4.

7.9 Error Expansion for Symmetric Langevin Methods

Of particular interest for sampling the canonical distribution are symmetric Langevin methods. We believe these are likely to be the most useful class of methods for practitioners, as by symmetrizing the expansion the odd order terms in (7.16) vanish identically using the Jacobi identity in the BCH expansions. This implies that a symmetric scheme gives a second order error in computed averages. Many symmetric methods can be constructed that require only one evaluation of the force per iteration (effectively making them as inexpensive as a first order method).

In order to simplify the algebra, we will use $\boldsymbol{M} = \boldsymbol{I}$ throughout this section, which will not alter the major results and amounts to a linear change of variables. We shall consider general symmetric schemes of the form $[\![\mathrm{XYZYX}]\!]$, and perform a similar analysis to the previous section. The terms appearing in the expansions will be significantly more complex than those encountered in first order methods. We make use of the following proposition for symmetric compositions:

Proposition 7.1 *Let $\mathcal{L}_{\mathrm{X}}^\dagger$, $\mathcal{L}_{\mathrm{Y}}^\dagger$ and $\mathcal{L}_{\mathrm{Z}}^\dagger$ be general operators to which BCH can be applied, then defining*

$$e^{h\hat{\mathcal{L}}^\dagger} = e^{\frac{h}{2}\mathcal{L}_{\mathrm{X}}^\dagger} e^{\frac{h}{2}\mathcal{L}_{\mathrm{Y}}^\dagger} e^{h\mathcal{L}_{\mathrm{Z}}^\dagger} e^{\frac{h}{2}\mathcal{L}_{\mathrm{Y}}^\dagger} e^{\frac{h}{2}\mathcal{L}_{\mathrm{X}}^\dagger}, \tag{7.26}$$

gives

$$\hat{\mathcal{L}}^\dagger = \mathcal{L}_0^\dagger + h^2 \mathcal{L}_2^\dagger + \mathcal{O}(h^4),$$

where

$$\begin{aligned}
\mathcal{L}_0^\dagger =& \mathcal{L}_{\mathrm{X}}^\dagger + \mathcal{L}_{\mathrm{Y}}^\dagger + \mathcal{L}_{\mathrm{Z}}^\dagger, \\
\mathcal{L}_2^\dagger =& \frac{1}{12}\left([\mathcal{L}_{\mathrm{Z}}^\dagger, \mathcal{L}_{\mathrm{Z}}^\dagger, \mathcal{L}_{\mathrm{Y}}^\dagger] + [\mathcal{L}_{\mathrm{Y}}^\dagger + \mathcal{L}_{\mathrm{Z}}^\dagger, \mathcal{L}_{\mathrm{Y}}^\dagger + \mathcal{L}_{\mathrm{Z}}^\dagger, \mathcal{L}_{\mathrm{X}}^\dagger] \right) \\
& - \frac{1}{24}\left([\mathcal{L}_{\mathrm{Y}}^\dagger, \mathcal{L}_{\mathrm{Y}}^\dagger, \mathcal{L}_{\mathrm{Z}}^\dagger] + [\mathcal{L}_{\mathrm{X}}^\dagger, \mathcal{L}_{\mathrm{X}}^\dagger, \mathcal{L}_{\mathrm{Y}}^\dagger + \mathcal{L}_{\mathrm{Z}}^\dagger] \right).
\end{aligned}$$